

Prompt Engineering

It is the discipline in which we design the input(prompt) in such a way that can guide LLM to give more accurate , reliable, and usefull outputs

Context Window

AI has a memory limit. The maximum tokens the AI model can remembers in once is called context window.

eg: suppose an model has context window of 8,000 tokens and you gave 10,000 tokens . So model will forget the initial 2000 tokens.

- ChatGPT : 128k to 256k
- Gemini : 1M-2M
- Cloude : 200K
- Perplexity : 200K

Components of Prompts

1. Role (Persona): We give AI an identity

eg: Act as a senior content manager

2. Context (Background) : Give background information about the task that you are giving

eg : I am launching a new organic coffee brand for health conscious people.

3. Task : Give clear task instructions

eg : You writes engaging blog posts with a 300 word limit.

4. Constraints: Tell AI what not to do

eg: Avoid using jargons

5. Format : Telling how the result should look like

eg: Output should be in markdown format

Core Prompting Techniques

1. Zero Shot Prompting (The Direct Order)

Zero shot prompting mean we do not provided any example (shot) to AI. We have only provided with instructions AI will provide answer on the basis of its pretrained knowledge.

```
Prompt: Classify the sentiment of this review:"The battery life of the laptop is good but screen is bit dull
```

```
Output: Sentiment: Mixed/Neutral
```

2. Few shot prompting (The pattern maker)

Sometimes the zero shot prompting fails specially when we want answer in specific format. Then we use few-shot prompting and provide AI with some examples so that model will learn some pattern.

```
Prompt: Classify the sentiment of this review; "The battery life of the laptop is good but screen is dull"
```

```
example1: "The delivery was late" -> {"sentiment": "negative", "label": 0}
```

```
example2: "I loved pizza" -> {"sentiment": "positive", "label": 1}
```

```
output: {"sentiment": "mixed", "label": 2}
```

3. Chain of thought (COT)

This is the important technique that is used to solve the complex problem. In this we ask AI to think step by step before giving the answer directly so that AI thinks logically .

```
Prompt: "There are 1,000 bottles of juice. Exactly one bottle is
poisoned. The poison is deadly but takes 24 hours to work.You have 10
test rats, and you can perform only one round of testing.How can you
identify the poisoned bottle within 24 hours ?Think step by step"
```

4. Instruction Prompting

In this we provide certain boundaries for model to perform the task. We use three things

- Persona (Role)
- Constraints (Limits)
- Specific Instructions

```
Role: Act as a Polite HR manager
```

```
Instructions: Write a rejection email for a candidate who cleared
the interview but we cant hire due to budget cuts.Mention that we will
keep CV for the future.
```

```
Constraints: Keep the word limit of 300 and be empathetic.
```

5. Json Prompting

In this we ask AI to produce answer in a specific Json Format

```
Prompt: You are a sentiment analysis assistant.
```

```
Analyze the following review and return the result in JSON.
```

```
Review:
```

```
"The battery lasts all day, but the screen is disappointing."
```

```
Output JSON format:
```

```
{
  "sentiment": "",
  "confidence": 0,
  "reason": ""
}
```

Advanced prompting Technique

1. ReAct

ReAct mean reasoning and acting. In this AI will work in loop : First it thoughts , perform action, then observe and then again thoughts and continues till final answers is achieved

Analyze the current stock performance of NVIDIA compared to its competitors in last 7 days

Instructions

- 1) start with a thought (What do you need to find)
- 2) perform action (Search for specific data)
- 3) provide observation (summarize what you find)
- 4) repeat untill you can provide a final "reccomendation" for a short term investment

2. Tree of thought:

Tree of thought can considered as updated version of Chain of thought. In ToT AI will explore like different branches of tree.

AI will generate 3-4 ideas . Evaluate those ideas themselves and find the next one.

Solve the following mathematics problem.

Instructions:

- Generate at least three possible solving methods.
- Estimate which method is most efficient.
- Discard inefficient approaches.
- Solve using the best approach.
- Verify the result using another method if possible.

Problem:

Find the maximum area of a rectangle inscribed in a semicircle of radius 10.